AquaSense Pro

Water Quality Prediction and Aquaculture IoT ML Project
Round 2 Prototype Submission Report

Project Focus

AquaSense Pro monitors water quality and purifier performance using live telemetry, batch CSV uploads, and simulati

1. Project Objective

This project develops an aquaculture water-quality and purifier monitoring system that combines IoT telemetry, sensor validation, and predictive analytics to improve operational visibility and decision support for end users.

The solution focuses on real-time monitoring of water parameters, process health, and purifier performance while remaining transparent about hardware limitations, unavailable channels, and simulated demo data.

The dashboard operates in three modes: live USB telemetry, uploaded CSV telemetry, and scenario-driven simulation. This helps maintain usability for testing, demonstration, and further hardware validation.

2. Problem Definition

Aquaculture operations depend on stable water quality, filter performance, and process continuity. Poor data quality or missing telemetry can lead to incorrect interpretation of system health and delayed intervention.

The project addresses this by validating sensor inputs, normalizing telemetry, and clearly identifying when channels are missing or invalid. It also warns the user when a sensor is missing, faulty, or out of range.

In live USB mode, the application waits for real readings and intentionally does not fabricate values. This preserves trust in the prototype and prevents false monitoring claims when the board is absent.

3. Functional Scope

The dashboard uses a segmented data-source interface for Live USB, Upload CSV, and Simulation. The live path checks for connected serial devices and reports a clear no-device state when no ESP32 is available.

The CSV path allows project users to upload telemetry, preview a bundled sample, or use a generated template. Simulation mode provides scenarios such as nominal operation, pressure trip, sensor fault, and flow imbalance.

Telemetry is normalized into metrics such as pH, temperature, turbidity, pressure, flow, TDS rejection, RO recovery, and mass-balance alarms. These outputs are displayed with plots and metrics only when the corresponding data channels are available.

4. Validation and Engineering Notes

The telemetry pipeline was tested with real device-compatible CSV content and simulated examples. The project documents that live hardware validation requires an actual connected ESP32, and the dashboard never claims to measure a physical process when the board is absent.

This is a critical design decision to preserve trust and avoid false readings. The app clearly warns the user when calibration or hardware validation remains pending, which is a responsible engineering approach for a prototype system.

The project includes tests for telemetry parsing, alarm generation, and sample-data handling. This provides a dependable baseline and makes the solution easier to extend toward future hardware integration and operational deployment.

Submission Summary

Project: AquaSense Pro
Domain: Water Quality Prediction and Aquaculture IoT ML
Primary interfaces: Live USB, CSV upload, and simulation dashboard
Status: Prototype completed with real telemetry handling and validation workflow
Key principle: The dashboard distinguishes clearly between measured hardware data and simulated data.
Prepared for review and further enhancement in the next prototype round.

Prepared for project review and prototype submission.